

Projective Signed-Permutation Classification of Exact Score Ties in the 2,285-State Plastic Closure

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Abstract

Let $\alpha > 1$ satisfy $\alpha^3 - \alpha - 1 = 0$. We classify every exact candidate-score equality on the 2,285 plastic-field replay states in version 9 of the finite-closure archive associated with Tagnon's complex cubic $\sin^2$ -type algorithm. For two distinct admissible moves M, N , including both V - V and V - W pairs, the scores are equal if and only if the image triples $M(s)$ and $N(s)$ differ by a common nonzero scalar in $\mathbb{Q}(\alpha)$, a coordinate permutation, and independent coordinate signs. The sufficient direction is a global identity. For the converse, the two field elements e_2/e_1^2 and e_3/e_1^3 , formed from the squared image coordinates, are a complete invariant of this projective signed-permutation relation; an exact finite computation then checks the converse on the archived domain. The calculation partitions 2,974,862 candidates and thereby covers 46,929,906,752 unordered pairs. It recovers exactly the archived 156 minimum-tie states and their tied digit sets. The theorem is confined to the specified replay representatives: it neither extends the biconditional to all plastic-field states nor reconstructs raw replay logs omitted from the archive.

1 Finite problem and result

Tagnon introduced a deterministic analytic continuation of Karpenkov's $\sin^2$ -algorithm to cubic fields of signature $(1, 1)$ [1, 2]. Exact score ties make a declared tie-breaking order necessary. In the plastic field, the accompanying finite closure contains 2,285 states and has an exact minimum tie at 156 of them. Open Problem 5 of [2] asks which moves tie, on which states, and why. We resolve precisely the frozen finite plastic layer of that question. The general tie problem, even elsewhere in the plastic field, remains outside the result.

Put

$$K = \mathbb{Q}(\alpha), \quad \alpha^3 = \alpha + 1,$$

and identify K with its real embedding σ_r when writing inequalities. A state is a $\mathbb{Q}$ -basis $s = (u_1, u_2, u_3)$ of K , with coordinates in $\mathbb{Z}[\alpha]$. Before candidate enumeration it is relabelled so that

$$x = \sigma_r(u_1) > y = \sigma_r(u_2) > z = \sigma_r(u_3) > 0.$$

The candidate matrices are

$$V(a, b, g) = \begin{pmatrix} 1 & -g & gb - a \\ 0 & 1 & -b \\ 0 & 0 & 1 \end{pmatrix}$$

for integers

$$0 \leq a \leq \lfloor x/z \rfloor, \quad 0 \leq b \leq \lfloor y/z \rfloor, \quad 0 \leq g \leq \left\lfloor \frac{x/z - a}{y/z - b} \right\rfloor,$$

excluding $(a, b, g) = (0, 0, 0)$, together with

$$W = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ -1 & 1 & 1 \end{pmatrix} \quad \text{when } z > x - y > 0.$$

These are the moves and admissibility windows of [1, 2].

For an image triple $U = (v_1, v_2, v_3)$, write

$$r = (\sigma_r(v_i))_{i=1}^3, \quad c = (\sigma_c(v_i))_{i=1}^3, \quad n_1 = r \times c, \quad n_2 = r \times \bar{c},$$

where dot and cross products are bilinear, not Hermitian. Its score is

$$q(U) = \operatorname{Re} \frac{\langle n_1 \times n_2, n_1 \times n_2 \rangle}{\langle n_1, n_1 \rangle \langle n_2, n_2 \rangle}. \quad (1)$$

The candidate with the most negative score is selected. The denominator is nonzero for a basis triple by the non-isotropy theorem of [3].

Define an equivalence relation on triples by

$$U \sim_{\text{sp}} U' \iff U' = \lambda P U \quad (2)$$

for some $\lambda \in K^\times$ and some signed permutation matrix P . Equivalently, after a permutation $\pi \in S_3$ there are independent $\varepsilon_i \in \{\pm 1\}$ such that $v'_i = \lambda \varepsilon_i v_{\pi(i)}$.

The finite domain $\mathcal{D}_{\text{pl},9}$ in the next theorem consists of the 2,285 rows of the archived plastic `closure_states` table, interpreted using the exact `verification_key` representative and `replay_mode` in the corresponding version-9 `exact_verification` table [4]. This convention is part of the statement, not an implementation detail.

Theorem 1 (Complete finite score-orbit classification). *For every $s \in \mathcal{D}_{\text{pl},9}$ and every two distinct admissible candidate matrices M, N at s ,*

$$q(Ms) = q(Ns) \iff Ms \sim_{\text{sp}} Ns.$$

This covers every V – V and V – W pair. Exactly 156 states have an equality at the minimum score, and their tied digit sets agree exactly with the archive: 126 are two-way and 30 are three-way ties.

The closest sources delimit the contribution sharply. The first paper reports the 2,974,862-candidate replay and the 156 minimum-tie states, but not an all-pair structural classification [2]. The companion proves dual, conformal, and non-isotropy identities and records particular ties, but does not give the biconditional in theorem 1 [3]. A focused search of those sources and current arXiv indexing on 7 September 2026 found no equivalent classification; this is a bounded literature assessment, not a priority claim.

2 The orbit invariant

We first prove the implication in theorem 1 that does not depend on the finite archive.

Proposition 2 (Global invariance). *Whenever the two scores are defined,*

$$U \sim_{\text{sp}} U' \implies q(U) = q(U').$$

Proof. Write $U' = \lambda PU$, where P is a signed permutation matrix. At the three embeddings,

$$r' = \sigma_r(\lambda)Pr, \quad c' = \sigma_c(\lambda)Pc, \quad \bar{c}' = \overline{\sigma_c(\lambda)}P\bar{c}.$$

For every orthogonal signed permutation matrix,

$$\langle Px, Py \rangle = \langle x, y \rangle, \quad (Px) \times (Py) = \det(P)P(x \times y).$$

Consequently the numerator and denominator of the quotient in (1) are both multiplied by

$$\sigma_r(\lambda)^4 (\sigma_c(\lambda) \overline{\sigma_c(\lambda)})^2.$$

Their quotient, and hence its real part, is unchanged. $\square$

For $U = (v_1, v_2, v_3)$ put $t_i = v_i^2$ and let

$$e_1 = t_1 + t_2 + t_3, \quad e_2 = t_1 t_2 + t_1 t_3 + t_2 t_3, \quad e_3 = t_1 t_2 t_3.$$

The real embedding of e_1 is a sum of three squares and is positive, so $e_1 \neq 0$. Define

$$I(U) = \frac{e_2}{e_1^2}, \quad J(U) = \frac{e_3}{e_1^3}. \quad (3)$$

Proposition 3 (Complete invariant). *If all coordinates of U and U' are nonzero, then*

$$U \sim_{\text{sp}} U' \iff (I(U), J(U)) = (I(U'), J(U')).$$

Every candidate image in theorem 1 satisfies the nonzero-coordinate hypothesis.

Proof. The forward implication follows because a common scalar multiplies the t_i by λ^2 , while signs disappear and a permutation only reorders them. Conversely, the three normalized squares t_i/e_1 are exactly the roots, with multiplicity, of

$$X^3 - X^2 + I(U)X - J(U). \quad (4)$$

Thus equality of the two invariants gives a permutation π for which

$$\frac{(v'_i)^2}{e_1(U')} = \frac{v_{\pi(i)}^2}{e_1(U)}.$$

Take $\lambda = v'_1/v_{\pi(1)}$. Every quotient $v'_i/v_{\pi(i)}$ then has square λ^2 ; since K has characteristic zero, it equals λ or $-\lambda$. This is (2). Finally, the rows of every candidate matrix are nonzero and the matrix is unimodular. A zero image coordinate would give a nontrivial rational linear relation among the basis elements of s , which is impossible. $\square$

The invariant is more than a convenient fingerprint: it turns the structural condition in theorem 1 into equality of two exact elements of a cubic field. In particular, once score equality is known in the finite calculation, the converse reduces to comparing (3), rather than searching for λ , P , and three signs.

3 A direct exact score formula

We next remove complex and floating-point arithmetic from the finite test. For $x = c_0 + c_1\alpha + c_2\alpha^2 \in K$, represent the chosen complex embedding as

$$\sigma_c(x) = \frac{A(x)}{2} + B(x)\mathcal{J}, \quad 4\mathcal{J}^2 = R := 4 - 3\alpha^2,$$

where

$$A(x) = 2c_0 + 2c_2 - c_1\alpha - c_2\alpha^2, \quad B(x) = c_1 - c_2\alpha.$$

For $U = (v_1, v_2, v_3)$ and cyclic pairs $(j, k) = (2, 3), (3, 1), (1, 2)$, set

$$P_i = v_j A(v_k) - v_k A(v_j), \quad S_i = v_j B(v_k) - v_k B(v_j),$$

and then

$$D = \sum_i (P_i^2 + S_i^2 R), \quad E = \sum_i P_i S_i, \quad H_i = S_j P_k - P_j S_k.$$

Lemma 4 (Exact plastic-field score). *For every candidate image,*

$$q(U) = \frac{4R \sum_i H_i^2}{D^2 - 4E^2 R} \in K. \quad (5)$$

Proof. The i th component of n_1 is $P_i/2 + S_i\mathcal{J}$, while n_2 is obtained by replacing $\mathcal{J}$ by $-\mathcal{J}$. Direct multiplication gives

$$\langle n_1, n_1 \rangle = D/4 + E\mathcal{J}, \quad (n_1 \times n_2)_i = H_i\mathcal{J}.$$

Substitution in (1), using $\mathcal{J}^2 = R/4$, yields (5). Every displayed operation is in $\mathbb{Q}[\alpha]/(\alpha^3 - \alpha - 1)$, and non-isotropy makes the denominator nonzero. $\square$

For completeness, division in (5) was canonicalized without numerical embeddings. If $z = a + b\alpha + c\alpha^2$, then

$$z^{-1} = N(z)^{-1}(A_z + B_z\alpha + C_z\alpha^2),$$

where

$$\begin{aligned} A_z &= a^2 + 2ac - b^2 - bc + c^2, & B_z &= -ab + c^2, & C_z &= -ac + b^2 - c^2, \\ N(z) &= a^3 + 2a^2c - ab^2 - 3abc \\ &\quad + ac^2 + b^3 - bc^2 + c^3. \end{aligned}$$

Multiplication modulo $\alpha^3 - \alpha - 1$ verifies $z(A_z + B_z\alpha + C_z\alpha^2) = N(z)$. Reducing the resulting rational coefficient quadruple to primitive form therefore gives an exact canonical key: equal keys are equivalent to equality in K .

As an illustration, consider the state with coordinate rows

$$(-1, -1, 2), \quad (-1, 1, 1), \quad (0, -1, 1).$$

The three minimum candidates $V(0, 1, 2), V(2, 2, 2), V(2, 2, 3)$ have the common values

$$I = \frac{3 - 2\alpha + \alpha^2}{7}, \quad J = \frac{-22 - 81\alpha + 79\alpha^2}{343}, \quad q = \frac{1 - 10\alpha - 7\alpha^2}{7}.$$

Their equality is therefore an instance of the orbit mechanism, not a floating-point coincidence.

4 Exhaustive finite certificate

For each archived representative, the computation decides the exact real sorting, all floor bounds, W admissibility, and all candidate images. It then evaluates (5) and partitions the candidates at that state by their canonical score keys. It is unnecessary to materialize a 46.9-billion-row pair table: every unequal-key pair is automatically unequal, and checking that each nonsingleton score class has one (I, J) key checks the score-equality-to-orbit implication for every equal pair. The reverse implication is already theorem 2.

The exact totals are shown in table 1.

Table 1: Complete counts on $\mathcal{D}_{\text{pl},9}$.

Quantity	Count
States	2,285
Admissible candidates	2,974,862
All unordered pairs	46,929,906,752
V – V pairs	46,929,854,476
V – W pairs	52,276
Nonsingleton exact-score classes	1,987
Exact equal-score pairs	2,541
Equal V – V pairs	2,505
Equal V – W pairs	36
States with some equality	482
States with a nonminimum equality	434
States with a minimum tie	156

There are 1,710 score classes of size two and 277 of size three, giving $1710 + 3 \cdot 277 = 2541$ equal pairs. Every one of the 1,987 classes has exactly one orbit key. Independently minimizing every state gives the same 156 tied states and the same tied digit sets as the archived tie table.

Two exact implementations checked the certificate. The first uses the archived candidate enumerator together with the independently derived direct formula (5). A second implementation imports neither that analyzer nor the archived scoring core: it uses integer-coordinate field arithmetic and a 220-bit rational isolating interval for the real root to certify signs and floors. It independently reproduces every total in table 1, both full-partition fingerprints, and every minimum-tie digit set, with no zero denominator. A separate formula check compares the direct expansion with the unchanged archived exact core on 687 candidates, including the displayed three-way minimum tie and a nonminimum V – W equality.

This certificate architecture is deliberately that of a finite-library theorem: exact predicates, signed-coordinate-permutation structure, independent enumeration, and an explicit warning against extrapolation. A closely related architecture is used for finite ternary Borsuk and kissing-code models in [5]; there signed coordinate permutations compress the certificate, whereas here the corresponding projective symmetry is itself the classified equality mechanism.

Proof of theorem 1. Proposition 2 proves orbit equivalence implies score equality for every defined score. Proposition 3 identifies orbit equivalence exactly with equality of (I, J) . The exhaustive exact partition just described checks that every nonsingleton score class on every one of the 2,285 frozen representatives has a single such invariant. This proves the converse for all 46,929,906,752 candidate pairs. The independent exact minimization and

comparison with the archived tie table give the final assertion about the 156 minimum-tie states and their multiplicities. $\square$

5 Scope, provenance, and reproducibility

The result is exact but finite. It does not prove the biconditional for every admissible state over K , for another cubic field, or for the HAPD tie events also mentioned in Open Problem 5. It classifies all score equalities on the specified table, including the many equalities away from the selected minimum; it does not make the lexicographic tie-break canonical.

There is one archive-level provenance limitation. Of the 2,285 rows, 482 use the archived `RAW_EACH_STEP` replay mode. The historical raw EXP28B flow logs needed to reconstruct those keys are absent, so the theorem takes their version-9 `verification_key` assignments as inputs. This does not introduce a numerical assumption. In the 466 rows where the archived source and verification strings differ, an exact transfer check shows that the two triples differ by a common positive-real field scalar and a signed permutation after normalization; their candidate windows, digit labels, and scores therefore correspond. Nevertheless, the omitted historical mapping is not claimed to have been independently reconstructed.

The source archive is available at [4]. The ancillary project files include the exact analyzer and serialized partition, the direct formula cross-check, and the independently written verifier. They use no floating-point tolerance to assert an equality. Version identifiers and file digests belong to the accompanying manifest rather than to the mathematical statement.

OpenAI Codex assisted with the structural search, exact-computation code, literature screening, adversarial checking, and manuscript preparation. Language-model output is not mathematical evidence. The proof is the displayed algebra together with the replayable exact finite certificates; the separate model context used for the second implementation is an audit pass, not independent peer review.

References

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